EXPERIMENT 1 – DESIGN STAR AND SNOWFLAKE SCHEMA

Aim

To design Star Schema and Snowflake Schema for the Bank Customer Churn system using a suitable fact table and dimension tables.

Objectives

- To understand the concept of Data Warehouse Schema.
- To design a Star Schema for the Bank Customer Churn dataset.
- To design a Snowflake Schema for the Bank Customer Churn dataset.
- To identify suitable fact and dimension tables.
- To understand the difference between Star Schema and Snowflake Schema.

Problem Statement : Bank Customer Churn

Banks maintain customer information such as credit score, geography, gender, age, account balance, tenure, products used, credit-card ownership, active membership and estimated salary. Some customers may leave the bank, which is known as customer churn. Organizing this information in a data warehouse helps the bank study customer characteristics and analyze churn patterns for better decision-making.

Theory

Data Warehouse Schema :-

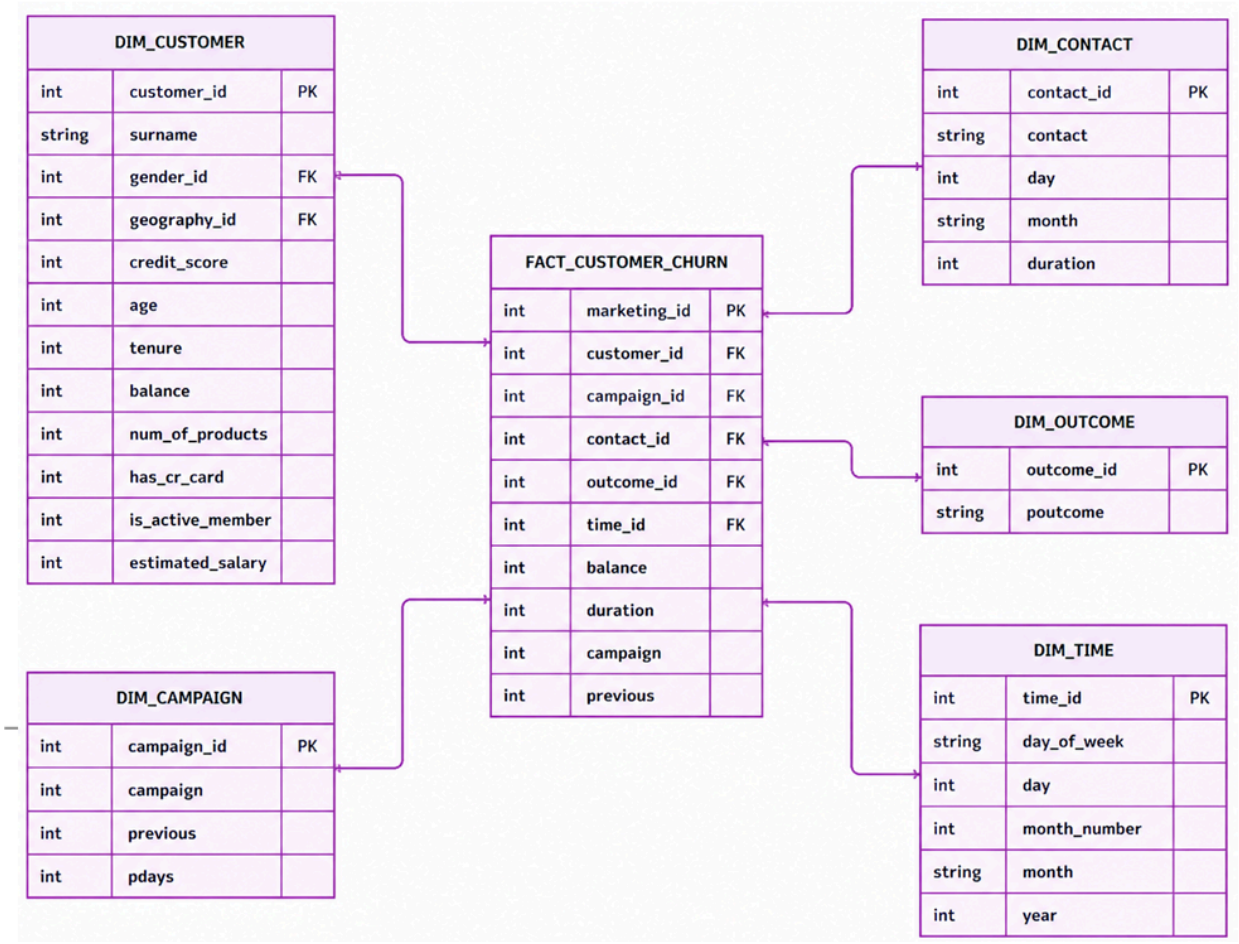
A Data Warehouse Schema defines how data is organized and related within a data warehouse. It mainly consists of Fact Tables and Dimension Tables. Star Schema and Snowflake Schema are two commonly used designs for organizing data for analysis.

1. Star Schema :-

A Star Schema is a data warehouse design in which a central fact table is directly connected to multiple dimension tables.

For the Bank Customer Churn system, FACT_CUSTOMER_CHURN is the central fact table. It stores customer measures such as Credit_Score, Age, Tenure, Balance, Num_Of_Products and Estimated_Salary, along with the Exited churn indicator. The dimension tables provide descriptive information about the customer, geography and gender.

Star Schema Diagram :-



Fact Table: FACT_CUSTOMER_CHURN

The fact table stores one record for each customer record in the source dataset.

Column	Description
Churn_Fact_ID	Primary Key
Customer_Key	Foreign Key
Geography_Key	Foreign Key
Gender_Key	Foreign Key
Credit_Score	Customer credit score
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
Num_Of_Products	Number of bank products used
Has_Cr_Card	Whether customer has a credit card
Is_Active_Member	Whether customer is an active member
Estimated_Salary	Estimated customer salary
Exited	Customer churn status (1 = left, 0 = stayed)

Dimension Tables

DIM_CUSTOMER

Column	Description
Customer_Key	Primary Key
CustomerId	Original customer identifier
Surname	Customer surname

DIM_GEOGRAPHY

Column	Description
Geography_Key	Primary Key
Geography	Customer's country / geographical category

DIM_GENDER

Column	Description
Gender_Key	Primary Key
Gender	Customer gender

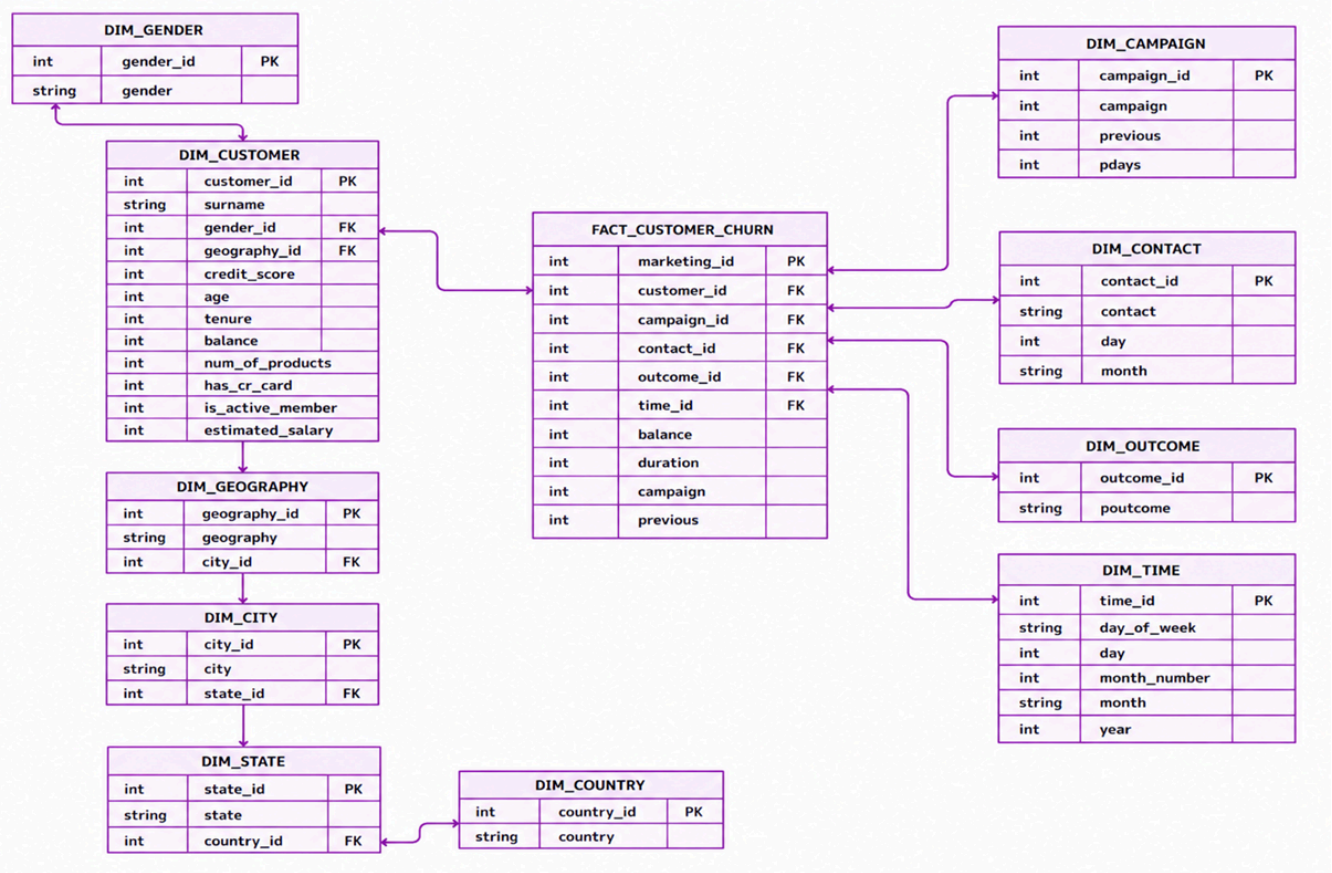
2. Snowflake Schema :-

A Snowflake Schema is a data warehouse schema that extends the Star Schema by normalizing some of its dimension tables into additional related tables. Instead of storing all descriptive information in a single dimension table, related information is separated into smaller tables.

For the Bank Customer Churn system, FACT_CUSTOMER_CHURN remains the central fact table. DIM_CUSTOMER stores customer details and contains foreign keys that refer to DIM_GEOGRAPHY and DIM_GENDER. These separate tables store the geography and gender categories.

The structure reduces repetition of descriptive information and makes the dimension tables more normalized. However, more table joins may be required when retrieving information.

Snowflake Schema Diagram :-



Fact Table: FACT_CUSTOMER_CHURN

The same fact table is used as the center of the Snowflake Schema.

Column	Description
Churn_Fact_ID	Primary Key
Customer_Key	Foreign Key
Geography_Key	Foreign Key
Gender_Key	Foreign Key
Credit_Score	Customer credit score
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
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DIM_GENDER

Column	Description
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Difference Between Star Schema and Snowflake Schema

Feature	Star Schema	Snowflake Schema
Structure	Simple and star-like	More complex and hierarchical
Normalization	Generally denormalized	More normalized
Dimension Tables	Directly connected to Fact Table	Dimensions may be divided into sub-dimensions
Data Redundancy	Comparatively higher	Comparatively lower
Joins	Requires fewer joins	Requires more joins

Result :-

The Star Schema and Snowflake Schema for the Bank Customer Churn system were successfully designed. FACT_CUSTOMER_CHURN was identified as the central fact table, and DIM_CUSTOMER, DIM_GEOGRAPHY and DIM_GENDER were identified as the dimension tables. The Snowflake Schema further normalizes the customer information by separating Geography and Gender into related dimension tables.